

Faculty Vita: Said Al-Hallaj

1. Education

B.Sc. Chemical Engineering, Jordan University of Science and Technology, Jordan, 1990
M.Sc. Chemical Engineering, Jordan University of Science and Technology, Jordan, 1994
PhD Chemical Engineering, Illinois Institute of Technology, Chicago, IL, USA, 1999

2. Academic Experience

9/09-now Visiting Research Professor, Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL
5/00-5/09 Research Professor, Department of Chemical and Biological Engineering, Coordinator of IIT Renewable Energy Programs, IIT, Chicago, IL
10/99-8/00 Senior Research Associate: Li-Battery/PEM fuel cell project at IIT, sponsor: U.S. Army Research Office
3/99-9/99 Associate Visiting Researcher, Department of Applied Chemistry, College of Engineering, Tohoku University, Sendai, Japan: thermal characterization and calorimetric studies for Li-battery research

3. Non-Academic Experience

6/14-now Founder/CEO NETenergy, Chicago, IL
3/22-now Chief Battery Scientist, Beam Global (NYSE: BEEM), (acquired AllCell)
9/08-7/20 Chairman/CEO, All Cell Technologies LLC, Chicago, IL

4. Certification or professional registrations

None

5. Current membership in professional organizations

Electrochemical Society

6. Honors and Awards

- Stanford Top 2% researchers in Chemical Engineering
- Top 10 papers at the MIT A+B 2024 Energy Symposium
- Illinois Tech Alumni Professional Achievement Award, 2022
- Advisory Board member, Armour College of Engineering, Illinois Tech
- Industry Advisory Board, Department of Mechanical and Industrial Engineering, UIC
- Shell GameChanger Accelerator Powered by NREL (GCxN) 2019
- Wells Fargo Innovation Incubator (IN2), NREL, 2016
- New Venture Challenge, 2nd place, Booth School of Business, University of Chicago, 2015
- Clean Energy Challenge, 1st place award, 2015
- Chicago Magazine 2012 Green Award
- Embassy of France, Young Entrepreneur Initiative Award (YEI), 2007

7. Selected service activities

- Course development: developed and teaching a graduate level course entitled “ChE 451 Renewable Energy Technologies” and co-authored course textbook “Hybrid Hydrogen Systems”
- Engineering and Entrepreneurship: developed and teaching a course entitled “ChE 427 Entrepreneurship in Chemical Engineering”
- Technology Commercialization: currently serving as Commercialization Fellow for UIC Office of Technology Management and successfully spun off two university-based tech start-up companies

8. Selected Publications and presentations

- 1- M Qasem, M Haddadin, Y Yassin, S Ratrou, C Chen, S Stoyanov, S Al-Hallaj, and M Krishnamurthy, “Real-Time Electrochemical Model-Based BMS Control for Mitigating Li-Plating and Extending Battery Life”, IEEE Transactions on Transportations Electrifications, Vol. 11, No. 3, June 2025
- 2- S Stoyanov, T G. Ritter, S Pappu, S Al-Hallaj, and R Shahbazian-Yassar, “Effect of Separator Type, Formation Cycle, and Fluoroethylene Carbonate on the long-Term Cycling Performance of Lithium” Metal Half-Cells, Chemistry Select 2025, 10, e202501481
- 3- Yi Zenga, R J Clark, Y Galazutdinova, A Odukamaiya, S Al-Hallaj, M M Farid, S Kaur, J Woods, “Open-cycle thermochemical energy storage for building space heating: Practical system configurations and effective energy density”, Applied Energy 376 (2024) 124218
- 4- Yana Galazutdinova 1, Ruby-Jean Clark 1, Said Al-Hallaj 1, Sumanjeet Kaur 2 and Mohammed Farid. “New Thermochemical Salt Hydrate System for Energy Storage in Buildings”, Energies 2024, 17, 5228. <https://doi.org/10.3390/en17205228>
- 5- M Qasem, S Stoyanov, S Ratrou, M Haddadin, Y Yassin, C Chen, S Al-Hallaj, and M Krishnamurthy, “Synthetic Data-Integrated Li-ion Battery Modeling for eVTOL Energy Systems”, IEEE Access, Vol. 12, 2024
- 6- R.J. Clark, G. Gholamibozanjani, J. Woods, S. Kaur, A. Odukamaiya, S. Al-Hallaj, M. Farid, “Experimental screening of salt hydrates for thermochemical energy storage for building heating application”, Journal of Energy Storage, 51, PP 104415, 2022
- 7- A Goyal, E. Kozubal, J. Woods, M. Nofal, S. Al-Hallaj, “Design and performance evaluation of a dual-circuit thermal energy storage module for air conditioners”, Applied Energy 292, 116843, 2021
- 8- Y. Galazutdinova, S. Ushak, M. Farid, S. Al-Hallaj, M. Grágeda, “Development of the inorganic composite phase change materials for passive thermal management of Li-ion batteries Application”, Journal of Power Sources 491, (2021) 229624
- 9- Y Galazutdinova, S Al-Hallaj, M Grágeda, S Ushak, “Development of the inorganic composite phase change materials for passive thermal management of Li-ion batteries: material characterization”, International Journal of Energy Research 44 (3), 2011-2022, 2020
- 10-
- 11- “Energy Storage for Smart Grid Applications,” Said Al-Hallaj, Keynote Speaker: 2nd International Conference on Smart Grid and Renewable Energy, November 19-21, 2019, Doha, Qatar
- 12- “Thermal Management and Safety of Li-ion Batteries,” Said Al-Hallaj, Invited Speaker: AIChE Midwest Regional Conference, 2019, UIC, Chicago, IL USA

- 13- “Thermal Management and Safety Concerns for Li-ion Batteries in Electric Mobility and Autonomous Vehicles Applications,’ Said Al-Hallaj, Keynote Speaker: IEEE Vehicle and Propulsion Conference, August 27-30, 2018, Chicago, IL, USA
- 14- Numerical modeling of transient heat transfer in a phase change composite thermal energy storage (PCC-TES) system for air conditioning applications, Aljehani, LC Nitsche, S Al-Hallaj, Applied Thermal Engineering 164, 114522, 2020
- 15- Development of the inorganic composite phase change materials for passive thermal management of Li-ion batteries: material characterization, Y Galazutdinova, S Al-Hallaj, M Grágeda, S Ushak, International Journal of Energy Research, 2019
- 16- Exploring the charge reactions in a Li–O₂ system with lithium oxide cathodes and nonaqueous electrolytes T Zhang, R Amine, X Bi, Y Qin, M Li, S Al-Hallaj, F Huo, J Lu, K Amine Journal of materials chemistry A 7 (26), 15615-15620, 2019
- 17- Liu, Y Yuan, J Wen, K He, X Bi, Y Guo, Nano Letters 17 (5), 2959-2966, 2017
- 18- Preventing thermal runaway propagation in lithium ion battery packs using a phase change composite material: An experimental study, S Wilke, B Schweitzer, S Khateeb, S Al-Hallaj, Journal of Power Sources 340, 51-59, 2017
- 19- Semi-Empirical Modeling of Capacity Fade: A Practical Approach for Battery Pack Manufacturers, S Wilke, B Schweitzer, S Khateeb, S Al-Hallaj, ECS Transactions 73 (1), 109-119, 2016
- 20- Modeling of compression curves of phase change graphite composites using Maxwell and Kelvin models, MF Alzoubi, S Khateeb, S Al-Hallaj, Journal of Composite Materials 50 (8), 1123-1135, 2016

9. Professional development activities

- Leadership: Board Member, Armour College of Engineering at Illinois Tech, and Industry Advisory Board for Mechanical Engineering at UIC
- Technology Commercialization: serving as Commercialization Fellow for UIC Office of Technology Management and successfully spun off two university-based tech start-ups.
- Engineering and Entrepreneurship: mentoring startups at University of Chicago, Resurgence/Polsky Center, UIC, Argonne National Lab, University of Auckland
- Jordan Solar EV projects: worked closely with the Jordan parliament and USAID to draft and pass legislation to exempt EVs from import tariffs to boost EV adoption and catalyze sales growth. Forbes magazine recently published an article stating that in 2024, 65% of new cars in Jordan were electric, which is the second highest share of EV sales in the world, after Norway, despite many obstacles.
- Community Service: member of Chicago-Amman Sister Cities Committee and founding and board member of Connect.Jo, a USAID funded initiative that focuses on connecting Jordanian diaspora professionals with researchers and startups from Jordan.